$G = SL_2(\mathbb{F}_q) \supset T' \cong \mu_{q+1} \subseteq \mathbb{F}_{q^2}^\times \hookrightarrow GL_2(\mathbb{F}_q)$

$\cup$

$B = \left\{\begin{pmatrix} a & b \\ 0 & a^{-1}\end{pmatrix}\right\} \supset T = \left\{\begin{pmatrix} a & \\ & a^{-1}\end{pmatrix}\right\}$

$\cup$

$U = \left\{\begin{pmatrix} 1 & b \\ 0 & 1\end{pmatrix}\right\}$ Recall, $|G| = (q-1)q(q+1)$, $q = p^f$

<u>Prop</u> $U$ is a Sylow $p$-subgroup.

Now say $\ell$ is a prime, let

$$S_\ell = \text{Sylow } \ell\text{-sub of } T,$$
$$S'_\ell = \text{'' \quad '' } T'.$$

<u>Prop</u> Let $\ell \neq p$ be prime dividing $|G|$.

1. If $\ell$ is odd and $\ell \mid (q-1)$, then $S_\ell$ is a Sylow $\ell$-subgroup of $G$. Moreover, $C_G(S_\ell) = T$ and $N_G(S_\ell) = N = \langle T, s\rangle$, $s = \begin{pmatrix} 0 & -1 \\ 1 & 0\end{pmatrix}$
2. If $\ell$ is odd and $\ell \mid (q+1)$, then $S'_\ell$ is a Sylow $\ell$-subgroup of $G$. Moreover, $C_G(S'_\ell) = T'$ and $N_G(S'_\ell) = N' = \langle T', s'\rangle$.
3. If $q \equiv 1 \pmod 4$ (resp. $q \equiv 3 \bmod 4$), then $\langle S_2, s\rangle$ (resp. $\langle S'_2, s'\rangle$) is a Sylow 2-subgroup of $G$.

<u>Proof</u> 1. If $\ell \mid (q-1)$ and $\ell$ is odd, $\ell \nmid (q+1)q$

$\Rightarrow S_\ell$ is a Sylow $\ell$-subgrp of $G$.

For $g \in S_\ell \setminus \{I_2\}$, we saw last time

$$C_G(g) = T \Rightarrow C_G(S_\ell) = T.$$

Since $\ell$ is odd and $\ell \mid (q-1)$, we have $q > 3$. So prop from last time $N_G(S_\ell)$ normalizes $C_G(S_\ell) = T$

$$\Rightarrow N_G(S_\ell) \subseteq N.$$

2. Similar.

3. If $q \equiv 1 \bmod 4$ (resp $q \equiv 3 \bmod 4$), then $4 \nmid (q+1)$ (resp. $4 \nmid (q-1)$), so

$$|\langle S_2, s\rangle| = 2|S_2| = \text{max power of 2 dividing } |G|$$

(resp. $|\langle S'_2, s'\rangle| = 2|S'_2| =$ '' ). $\square$

——//——

## <u>The Drinfeld curve</u>

Let $\mathbb{F}$ = an algebraic closure of $\mathbb{F}_q$.

The Drinfeld curve is the affine plane curve

$$Y = \{(x,y) \in \underset{\mathbb{F}^2}{\overset{\|}{\mathbb{A}^2(\mathbb{F})}} : xy^q - yx^q = 1\}$$

Then

- The action of $G = SL_2(\mathbb{F}_q)$ on $\mathbb{A}^2(\mathbb{F})$ by $\begin{pmatrix} a & b \\ c & d\end{pmatrix}(x,y) = (ax+by, cx+dy)$ preserves $Y$.

  One way to think of this: $G$ preserves the symplectic pairing $\langle (x,y),(u,v)\rangle = xv - uy$ on $\mathbb{F}^2$

  $\Rightarrow G$ preserves the fibres of

  $$\mathbb{A}^2(\mathbb{F}) \to \mathbb{A}^1(\mathbb{F})$$
  $$(x,y) \mapsto \langle (x,y), (x^q, y^q)\rangle$$

- $\mu_{q+1}$ on $\mathbb{A}^2(\mathbb{F})$ by $\alpha\cdot(x,y) = (\alpha x, \alpha y)$ preserves $Y$.

  Note, this is different from ~~our~~ the action of $\mu_{q+1} \cong T' \subseteq G$.

- The Frobenius endomorphism

  $$F : \mathbb{A}^2(\mathbb{F}) \to \mathbb{A}^2(\mathbb{F})$$
  $$(x,y) \mapsto (x^q, y^q)$$

  preserves $Y$.

So $Y$ has a lot of auto/endomorphisms. They satisfy, for $g \in G$, $\alpha \in \mu_{q+1}$,

$$g \circ \alpha = \alpha \circ g$$
$$g \circ F = F \circ g$$
$$F \circ \alpha = \alpha^q \circ F = \alpha^{-1} \circ F$$

<u>Later goal</u>: Compute $\ell$-adic cohomology of $Y$ with its actions of $G$, $\mu_{q+1}$, and $F$.

<u>Prop</u> $Y$ is affine smooth and irreducible.

<u>Proof</u> Clearly $Y$ is affine. To show it is irreducible, consider the automorphism of

$$\mathbb{A}^2(\mathbb{F}) \setminus \{y=0\} \to \mathbb{A}^2(\mathbb{F}) \setminus \{y=0\}$$
$$(x,y) \longmapsto \left(\frac{x}{y}, \frac{1}{y}\right) = (u,v)$$

This sends $Y$ to $\{v^{q+1} - u^q - u = 0\}$

and $v^{q+1} - u^q - u \in \mathbb{F}[u][v]$ is irred by Eisenstein's criterion.

To see $Y$ is smooth, we use the Jacobian criterion.

The Jacobian,

$$(y^q \quad -x^q) \text{ is } 0 \text{ only at } (0,0) \notin Y. \qquad \square$$

Prop $G$ acts freely on $Y$.

Proof Say $g \in G$ and $(x,y) \in Y$ satisfy

$$g(x,y) = (x,y)$$

So $g$ has $1$ as an eigenvalue. Then $g$ is conjugate in $G$ to $\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$ for some $b \in \mathbb{F}_q$, and changing $(x,y)$ by another point in its $G$-orbit, we can assume $g = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$.

Then $\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}(x,y) = (x,y) \iff x + by = x$.

But $(x,y) \in Y$, $y \neq 0 \Rightarrow b = 0$ and $g = I_2$. $\square$